



# United International University

## Department of Computer Science and Engineering

CSE 4883: Digital Image Processing Mid: Spring 2026

Total Marks: 20 Time: 1 hour and 30 minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all the questions. The numbers on the right of the questions denote their marks.

1. A rescue team is using a thermal drone to find a hiker stranded in a snowy, rocky landscape at dusk. The hiker is wearing a grey thermal blanket that is only  $1.5^{\circ}\text{C}$  warmer than the surrounding rocks. The hiker is likely curled up, occupying an area of about  $0.5\text{m} \times 0.5\text{m}$ . (3)
- Sensor X: Has a spatial resolution of **10cm/pixel** but a thermal sensitivity (radiometric depth) that groups temperatures into  $2^{\circ}\text{C}$  increments.
  - Sensor Y: Has a spatial resolution of **80cm/pixel** but a thermal sensitivity that distinguishes temperatures in  $0.1^{\circ}\text{C}$  increments.
- (a) Explain why Sensor X would likely fail to "see" the hiker even if the drone flies directly over them.
- (b) Explain why Sensor Y would likely fail to identify the hiker as a human-shaped object.
2. A deep-space probe has captured a high-resolution grayscale image of a crater on the Moon's dark side. The probe's transmitter is low on power, and a solar storm is approaching, meaning the image must be compressed and sent within a tiny data window. The onboard computer generates a Pixel Intensity Histogram of the crater floor to identify statistical redundancies before applying Huffman Coding. The distribution for a  $16 \times 12$  pixel block (**grayscale**) is as follows:

Table 1: Pixel Intensity Distribution for Satellite Crater Imagery

| Gray Level (Intensity)       | Appearance in the Image (Frequency) |
|------------------------------|-------------------------------------|
| 10 (Deep Crater Shadow)      | 42                                  |
| 55 (Volcanic Glass)          | 25                                  |
| 85 (Basaltic Rock)           | 28                                  |
| 120 (Regolith/Dust)          | 35                                  |
| 160 (Highland Slopes)        | 21                                  |
| 205 (Direct Sunlight)        | 9                                   |
| 240 (Ice/Reflective Mineral) | 15                                  |
| 255 (White light)            | 17                                  |

- (a) Remove the coding redundancy of this image. Draw the **Huffman tree** and build the **Huffman table** showing the memory requirement. (**You do not need to show every step of the tree build, but you must show the final tree**) (3)
- (b) After calculating the final image size, determine how much the image size has been reduced. Also, show the percentage of repeated data and the final intensity resolution requirements (bits/pixel). (2)
3. Consider the following **5x5 grayscale image** where  $V = [50-120, 155-210]$  (4)

|     |     |     |     |     |
|-----|-----|-----|-----|-----|
| 102 | 125 | 188 | 65  | 92  |
| 175 | 99  | 238 | 105 | 101 |
| 225 | 19  | 185 | 28  | 110 |
| 103 | 96  | 205 | 100 | 132 |
| 230 | 125 | 118 | 69  | 35  |

- (a) Find out and show the connectivity of all the 4-adjacent pixels.
- (b) Find out and show the connectivity of all the 8-adjacent pixels.
- (c) Find out and show the connectivity of all the m-adjacent pixels.

**You must draw the grid and show the connectivity for each type of adjacency.**

4. (a) Consider the following ANN: (The Input, Output, and weight matrices for different layers are also given)

(4)

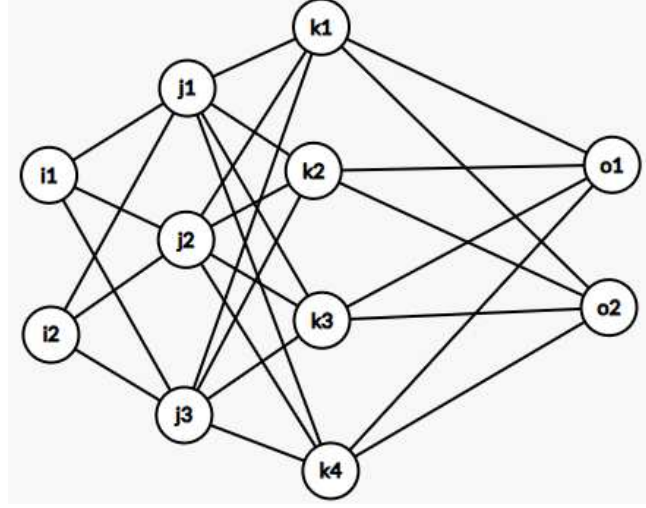


Figure 1: A sample ANN

$$\text{Inputs} = [1.35 \quad 2.65], \quad \text{Actual Outputs} = [1.0 \quad 0.0]$$

Weight of the 1st layer,

$$W_{ih1} = \begin{bmatrix} 0.35 & 0.45 & 0.20 \\ 0.25 & 0.60 & 0.15 \end{bmatrix}$$

Weight of the 2nd layer,

$$W_{h1h2} = \begin{bmatrix} 0.1 & 0.15 & 0.4 & 0.35 \\ 0.35 & 0.10 & 0.25 & 0.30 \\ 0.1 & 0.2 & 0.55 & 0.15 \end{bmatrix}$$

Weight of the 3rd layer,

$$W_{h2o} = \begin{bmatrix} 0.3 & 0.7 \\ 0.55 & 0.45 \\ 0.25 & 0.75 \\ 0.85 & 0.15 \end{bmatrix}$$

**The ANN output will be used to compare the likelihood of all the output categories.** Activation functions for the layers: **Sigmoid(layer 1)**, **ReLU(layer 2)**, and **suitable(layer 3/output layer)** based on the use-case.

- Now, for the given weights and input, determine the output of the ANN. (No need to show detailed calculation, but you must show the outputs of each layer. You must mention the choice of your activation function for the output layer.)
- Calculate the total loss using the log loss function given as follows:

$$\text{Log Loss} = -y_i \log(o_{outi}) - (1 - y_i) \log(1 - o_{outi})$$

(b) A laboratory uses a high-resolution scanner to analyze tissue samples represented as  $60 \times 60$  pixel grayscale images. To capture different levels of detail, the system uses a variable-grid feature extraction method:

(4)

- **The Outer Regions:** The first 3 rows and the last 3 rows of the image are processed using  $3 \times 3$  non-overlapping blocks.
- **The Inner Region:** The remaining middle rows are processed using  $6 \times 6$  non-overlapping blocks.
- **Feature Extraction:** From every single block (regardless of size), the system extracts **5** distinct statistical features.

The system identifies whether a sample contains one of the **4** specific aggressive cancer subtypes (Subtype A, B, C, or D) or if the tissue is Benign. However, the medical protocol only requires the model to trigger a single "Refer to Specialist" flag if any of the 4 cancer subtypes are detected, and a "Clear" flag if the tissue is Benign.

Based on this scenario,

- Show the **step-by-step** calculation to determine how many neurons are required for the input layer.
- Based on the clinical requirement described, how many neurons should be used in the output layer?
- Identify the most suitable activation function for the output layer and justify your choice based on the final goal of the system.